

Econometrics II

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Problem Set 8

Due: April 22, 2026

Question 1. Logit estimation via MLE, NLLS, and MoM

Estimate a logit model of the 4-year college attendance choice (relative to 2-year college) among post-secondary attendees using `dataHW8_Problem1.mat`. The outcome Y_i indicates attendance at a 4-year college. X_i contains individual characteristics including a constant, a parental college-graduation indicator, and the student's high-school GPA. Z contains choice-specific attributes: distance to the nearest 2-year college and distance to the nearest 4-year college (coefficients common across choices).

Solution. Because all three methods return the same estimates, I will report one table with the β estimates, γ estimates, and standard errors.

4-Yr College	Estimate	Std. Error
β_0	-2.05	0.19
β_1	1.30	0.05
β_2	0.65	0.06
γ_1	0.02	0.00

Table 1: Logit estimates of 4-year college attendance choice.

Question 2. Two-Period Labor Supply Model with Savings

Use `dataHW8_Problem2.mat`. Individuals are endowed with first-period savings drawn from $k_1 \sim \text{Uniform}[1.5, 3.5]$. In each period they jointly choose whether to work and how much to consume/save. Wages take five values $\{0.5, 0.75, 1.0, 1.5, 2.0\}$. The initial wage distribution is $(\delta_1, \delta_2, \delta_3, \delta_4, 1 - \sum_j \delta_j)$ and wages transition according to

$$\Pi = \begin{pmatrix} \rho_1 + \rho_2 + \rho_3 & \rho_2 & \rho_3 & 0 & 0 \\ \rho_2 + \rho_3 & \rho_1 & \rho_2 & \rho_3 & 0 \\ \rho_3 & \rho_2 & \rho_1 & \rho_2 & \rho_3 \\ 0 & \rho_3 & \rho_2 & \rho_1 & \rho_2 + \rho_3 \\ 0 & 0 & \rho_3 & \rho_2 & \rho_1 + \rho_2 + \rho_3 \end{pmatrix}$$

with $\rho_1 + 2\rho_2 + 2\rho_3 = 1$. Flow utility is

$$u(c_t, l_t) = \beta \ln c_t - \gamma \cdot \mathbf{1}(l_t = 1) + \varepsilon_{l_t}, \quad \varepsilon_{l_t} \stackrel{\text{iid}}{\sim} \text{Type I EV},$$

with discount factor $\delta = 0.9$. True parameter values: $\delta_j = 0.2$ for all j , $\rho_1 = 0.4$, $\rho_2 = 0.2$, $\rho_3 = 0.1$, $\beta = 2$, $\gamma = 0.5$.

(a) Construct a set of moments to target in estimation from `dataHW8_Problem2.mat`. Report the sample moments.

Solution.

Table 2: Sample Moments

Moment	Value
Mean Assets, $t = 1$	2.5029
Mean Assets, $t = 2$	1.2282
Std. Assets, $t = 2$	0.4081
Mean Savings ($k_2 - k_1$)	-1.2747
LFP, $t = 1$	0.6507
LFP, $t = 2$	0.6857
Share $w = 0.50$, $t = 1$	0.0000
Share $w = 0.75$, $t = 1$	0.0000
Share $w = 1.00$, $t = 1$	0.1657
Share $w = 1.50$, $t = 1$	0.0000
Share $w = 2.00$, $t = 1$	0.1875
Share $w = 0.50$, $t = 2$	0.0000
Share $w = 0.75$, $t = 2$	0.0000
Share $w = 1.00$, $t = 2$	0.1807
Share $w = 1.50$, $t = 2$	0.0000
Share $w = 2.00$, $t = 2$	0.1625
$\Pr(L_2 = 1 \mid w_1 \geq 1.50)$	0.6814
$\Pr(L_2 = 1 \mid w_1 \leq 0.75)$	0.7320
$\Pr(L_2 = 1 \mid L_1 = 1)$	0.6608
$\Pr(L_2 = 1 \mid L_1 = 0)$	0.7320
$\text{Corr}(k_1, L_1)$	-0.0810

(b) Simulate $N = 20,000$ observations at the true parameter values and verify that the simulated moments are close to the sample moments.

Solution.

Table 3: Data Moments vs. Simulated Moments at True Parameters

Moment	Data	Simulated	Difference
Mean Assets, $t = 1$	2.5029	2.4962	0.0067
Mean Assets, $t = 2$	1.2282	1.2474	-0.0191
Std. Assets, $t = 2$	0.4081	0.4086	-0.0005
Mean Savings ($k_2 - k_1$)	-1.2747	-1.2489	-0.0258
LFP, $t = 1$	0.6507	0.6128	0.0379
LFP, $t = 2$	0.6857	0.6314	0.0544
Share $w = 0.50$, $t = 1$	0.0000	0.1707	-0.1707
Share $w = 0.75$, $t = 1$	0.0000	0.1874	-0.1874
Share $w = 1.00$, $t = 1$	0.1657	0.2026	-0.0369
Share $w = 1.50$, $t = 1$	0.0000	0.2059	-0.2059
Share $w = 2.00$, $t = 1$	0.1875	0.2334	-0.0459
Share $w = 0.50$, $t = 2$	0.0000	0.1886	-0.1886
Share $w = 0.75$, $t = 2$	0.0000	0.1700	-0.1700
Share $w = 1.00$, $t = 2$	0.1807	0.1990	-0.0183
Share $w = 1.50$, $t = 2$	0.0000	0.1901	-0.1901
Share $w = 2.00$, $t = 2$	0.1625	0.2523	-0.0898
$\Pr(L_2 = 1 \mid w_1 \geq 1.50)$	0.6814	0.6850	-0.0036
$\Pr(L_2 = 1 \mid w_1 \leq 0.75)$	0.7320	0.5975	0.1346
$\Pr(L_2 = 1 \mid L_1 = 1)$	0.6608	0.6685	-0.0077
$\Pr(L_2 = 1 \mid L_1 = 0)$	0.7320	0.5726	0.1595
$\text{Corr}(k_1, L_1)$	-0.0810	-0.0518	-0.0292

(c) Estimate the model parameters by embedding the simulation inside an SMM estimator. Restrict the wage distribution probabilities and transition parameters to sum to one using a multinomial-logit-style reparameterization. Report estimates of β , γ , δ_j for $j = 1, \dots, 5$, and ρ_1, ρ_2, ρ_3 .

Solution.

Table 4: SMM Parameter Estimates

Parameter	True	Estimate
<i>Utility parameters</i>		
β	2.0	2.0702
γ	0.5	0.5000
<i>Wage transition</i>		
ρ_1	0.4	0.4001
ρ_2	0.2	0.1999
ρ_3	0.1	0.1000
<i>Initial wage distribution (δ_j)</i>		
$w = 0.50$	0.2	0.2000
$w = 0.75$	0.2	0.2000
$w = 1.00$	0.2	0.2000
$w = 1.50$	0.2	0.2000
$w = 2.00$	0.2	0.2000